

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

AUTOMATIC TEMPERATURE CONTROL MODULE (ATCM) - HYBRID ELECTRIC VEHICLE

WARNINGS:

- A risk assessment must be performed before any work is undertaken.
- All hybrid electric vehicle work must be performed by a suitably qualified person.
- Appropriate Personal Protection Equipment (PPE) must be worn when working on or near a hybrid electric vehicle high voltage system.
- The hybrid electric vehicle battery pack casing must not be opened due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- The electric power inverter converter casing must not be disassembled due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- If the hybrid electric vehicle battery pack is damaged or overcharged, there is a risk of exposure to hazardous voltage and/or highly corrosive electrolyte mist. If liquid or vapour is observed leaking from the hybrid electric vehicle battery pack,

take the following action:

- Evacuate the area
- Notify manager
- Do not breathe smoke/vapour
- Contain spillage using spill kit
- Wash any spillage off body and clothing (remove contaminated clothing)
- If the hybrid electric vehicle warning indicator is illuminated, the Battery Energy Control Module (BECM) cannot isolate the high voltage cables. Live working will be necessary to rectify the fault.
- The motor generator can generate a voltage if the wheels are rotated, even if the hybrid electric vehicle system has been made safe.

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Automatic Temperature Control Module (ATCM) - Hybrid Electric Vehicle. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B102A-11	Flame Sensor - Circuit short to ground	Fuel fired booster heater flame sensor circuit short circuit to ground	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault - Refer to the electrical circuit diagrams and check the flame sensor circuit for short circuit to ground. Repair the circuit or install a new flame sensor located within the fuel fired booster heater as necessary - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B102A-15	Flame Sensor - Circuit short to battery or open	Fuel fired booster heater flame sensor circuit short	

		circuit to power, open circuit, high resistance	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault - Refer to the electrical circuit diagrams and check the flame sensor circuit for short circuit to power, open circuit, high resistance. Repair the circuit or install a new flame sensor located within the fuel fired booster heater as necessary - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B102A-64	Flame Sensor - Signal plausibility failure	Fuel fired booster heater flame sensor circuit short circuit to ground, short	

		circuit to power, open circuit, high resistance	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault - Refer to the electrical circuit diagrams and check the flame sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the circuit or install a new flame sensor located within the fuel fired booster heater as necessary - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B102E-11	Air Quality Sensor - Circuit short to ground	Pollution sensor signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the pollution sensor circuit for short circuit to ground

B102E-15	Air Quality Sensor - Circuit short to battery or open	▪ Pollution sensor signal circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the pollution sensor circuit for short circuit to power, open circuit, high resistance
B102E-96	Air Quality Sensor - Component internal failure	▪ Pollution sensor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new pollution sensor
B1030-01	Left Front Seat Heater - General electrical failure	▪ Front left heated seat cushion element power or ground circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the front left heated seat cushion element power and ground circuits for open circuit, high resistance
B1030-4B	Left Front Seat Heater - Over temperature	▪ Front left heated seat cushion element internal failure ▪ Front left heated seat backrest element internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left heated seat cushion element ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left heated seat backrest element
B1030-87	Left Front Seat Heater - Missing message	▪ Front left heated seat cushion element power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the front left heated seat cushion element power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer

		▪ Front left heated seat cushion element internal failure ▪ Front left heated seat backrest element internal failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left heated seat cushion element ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left heated seat backrest element
B1031-01	Left Rear Seat Heater - General electrical failure	▪ Rear left heated seat cushion element power or ground circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear left heated seat cushion element power and ground circuits for open circuit, high resistance
B1031-4B	Left Rear Seat Heater - Over temperature	▪ Rear left heated seat cushion element internal failure ▪ Rear left heated seat backrest element internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left heated seat cushion element ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left heated seat backrest element
B1031-87	Left Rear Seat Heater - Missing message	▪ Rear left heated seat cushion element power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear left heated seat cushion element internal failure	▪ Refer to the electrical circuit diagrams and check the rear left heated seat cushion element power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a

		▪ Rear left heated seat backrest element internal failure	new rear left heated seat cushion element ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left heated seat backrest element
B1032-01	Right Front Seat Heater - General electrical failure	▪ Front right heated seat cushion element power or ground circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the front right heated seat cushion element power and ground circuits for open circuit, high resistance
B1032-4B	Right Front Seat Heater - Over temperature	▪ Front right heated seat cushion element internal failure ▪ Front right heated seat backrest element internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right heated seat cushion element ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right heated seat backrest element
B1032-87	Right Front Seat Heater - Missing message	▪ Front right heated seat cushion element power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Front right heated seat cushion element internal failure ▪ Front right	▪ Refer to the electrical circuit diagrams and check the front right heated seat cushion element power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right heated seat cushion element

		heated seat backrest element internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right heated seat backrest element
B1033-01	Right Rear Seat Heater - General electrical failure	Rear right heated seat cushion element power or ground circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear right heated seat cushion element power and ground circuits for open circuit, high resistance
B1033-4B	Right Rear Seat Heater - Over temperature	- Rear right heated seat cushion element internal failure - Rear right heated seat backrest element internal failure	- Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right heated seat cushion element - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right heated seat backrest element
B1033-87	Right Rear Seat Heater - Missing message	- Rear right heated seat cushion element power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear right heated seat cushion element internal failure - Rear right heated seat backrest	- Refer to the electrical circuit diagrams and check the rear right heated seat cushion element power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right heated seat cushion element - Using the manufacturer approved diagnostic system,

		element internal failure	clear the DTCs and retest. If the fault persists, install a new rear right heated seat backrest element
B1034-01	Left Front Seat Heater Element - General electrical failure	- Front left heated seat cushion element circuit open circuit, high resistance - Front left heated seat backrest element circuit open circuit, high resistance	- Refer to the electrical circuit diagrams and check the front left heated seat cushion element circuit for open circuit, high resistance - Refer to the electrical circuit diagrams and check the front left heated seat backrest element circuit for open circuit, high resistance
B1035-01	Left Rear Seat Heater Element - General electrical failure	- Rear left heated seat cushion element circuit open circuit, high resistance - Rear left heated seat backrest element circuit open circuit, high resistance	- Refer to the electrical circuit diagrams and check the rear left heated seat cushion element circuit for open circuit, high resistance - Refer to the electrical circuit diagrams and check the rear left heated seat backrest element circuit for open circuit, high resistance
B1036-01	Right Front Seat Heater Element - General electrical failure	- Front right heated seat cushion element circuit open circuit, high resistance - Front right heated seat backrest element circuit open circuit, high resistance	- Refer to the electrical circuit diagrams and check the front right heated seat cushion element circuit for open circuit, high resistance - Refer to the electrical circuit diagrams and check the front right heated seat backrest element circuit for open circuit, high resistance
B1037-01	Right Rear Seat Heater Element - General electrical failure	Rear right heated seat cushion element circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear right heated seat cushion element circuit for open circuit, high resistance

		▪ Rear right heated seat backrest element circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right heated seat backrest element circuit for open circuit, high resistance
B1038-01	Left Front Seat Heater Sensor - General electrical failure	▪ Front left heated seat cushion element internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left heated seat cushion element
B1039-01	Left Rear Seat Heater Sensor - General electrical failure	▪ Rear left heated seat cushion element internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left heated seat cushion element
B103A-01	Right Front Seat Heater Sensor - General electrical failure	▪ Front right heated seat cushion element internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right heated seat cushion element
B103B-01	Right Rear Seat Heater Sensor - General electrical failure	▪ Rear right heated seat cushion element internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right heated seat cushion element
B103F-87	Rear Seat Heater Control - Missing message	▪ Rear seat heater switchpack power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear seat heater switchpack power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		▪ Rear seat heater switchpack internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear seat heater switchpack
B105A-11	Cabin Temperature Sensor Fan - Circuit short to ground	▪ In-vehicle temperature sensor fan circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the in-vehicle temperature sensor fan circuit for short circuit to ground
B105A-12	Cabin Temperature Sensor Fan - Circuit short to battery	▪ In-vehicle temperature sensor fan circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the in-vehicle temperature sensor fan circuit for short circuit to power
B105A-13	Cabin Temperature Sensor Fan - Circuit open	▪ In-vehicle temperature sensor fan circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the in-vehicle temperature sensor fan circuit for open circuit, high resistance
B105A-97	Cabin Temperature Sensor Fan - Component or system operation obstructed or blocked	▪ In-vehicle temperature sensor fan jammed or broken	▪ Remove any obstructions from the fan. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new in-vehicle temperature sensor
B1083-49	Recirculation Damper Motor - Internal electronic failure	▪ Recirculation motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new recirculation motor
B1083-87	Recirculation Damper Motor - Missing message	▪ Recirculation motor power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to	▪ Refer to the electrical circuit diagrams and check the recirculation motor power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN

		ground, short circuit to power, open circuit, high resistance ▪ Recirculation motor internal failure	circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new recirculation motor
B1083-97	Recirculation Damper Motor - Component or system operation obstructed or blocked	▪ Recirculation motor jammed or broken	▪ Remove any obstructions from the recirculation motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Fresh/Recirculated Air Flap Actuator Requested Position (0x985A) - and - Fresh/Recirculated Air Flap Actuator Measured Position (0x9859). If inconsistent, install a new recirculation motor
B1085-49	Defroster Damper Motor - Internal electronic failure	▪ Demist distribution motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new demist distribution motor
B1085-87	Defroster Damper Motor - Missing message	▪ Demist distribution motor power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Demist distribution motor internal	▪ Refer to the electrical circuit diagrams and check the demist distribution motor power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If

		failure	the fault persists, install a new demist distribution motor
B1085-97	Defroster Damper Motor - Component or system operation obstructed or blocked	Demist distribution motor jammed or broken	Remove any obstructions from the demist distribution motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Defrost Air Flap Actuator Requested Position (0x9862) - and - Defrost Air Flap Actuator Measured Position (0x9861). If inconsistent, install a new demist distribution motor
B1087-4A	LIN Bus "A" - Incorrect component installed	Incorrect component connected to the LIN bus	Using the manufacturer approved diagnostic system, check for other LIN related DTCs and perform the relevant corrective action
B1087-88	LIN Bus "A" - Bus off	LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B1088-4A	LIN Bus "B" - Incorrect component installed	Incorrect component connected to the LIN bus	Using the manufacturer approved diagnostic system, check for other LIN related DTCs and perform the relevant corrective action
B1088-88	LIN Bus "B" - Bus off	LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B10AF-11	Blower Fan Relay - Circuit short to ground	Blower relay circuit short circuit to ground	Refer to the electrical circuit diagrams and check the blower relay circuit for short

			circuit to ground
B10AF-12	Blower Fan Relay - Circuit short to battery	Blower relay circuit short circuit to power	Refer to the electrical circuit diagrams and check the blower relay circuit for short circuit to power
B10AF-13	Blower Fan Relay - Circuit open	Blower relay circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the blower relay circuit for open circuit, high resistance
B10B0-12	Rear Blower Fan Relay - Circuit short to battery	Rear climate control blower relay circuit short circuit to power	Refer to the electrical circuit diagrams and check the rear climate control blower relay circuit for short circuit to power
B10B0-13	Rear Blower Fan Relay - Circuit open	Rear climate control blower relay circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear climate control blower relay circuit for open circuit, high resistance
B10B0-14	Rear Blower Fan Relay - Circuit short to ground or open	Rear climate control blower relay circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear climate control blower relay circuit for short circuit to ground, open circuit, high resistance
B10B9-11	Blower Control - Circuit short to ground	Blower control circuit short circuit to ground	Refer to the electrical circuit diagrams and check the blower control circuit for short circuit to ground
B10B9-12	Blower Control - Circuit short to battery	Blower control circuit short circuit to power	Refer to the electrical circuit diagrams and check the blower control circuit for short circuit to power
B10B9-13	Blower Control - Circuit open	Blower control circuit open circuit, high	Refer to the electrical circuit diagrams and check the blower control circuit for

		resistance	open circuit, high resistance
B10BA-11	Rear Blower Control - Circuit short to ground	▪ Rear climate control blower control circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear climate control blower control circuit for short circuit to ground
B10BA-12	Rear Blower Control - Circuit short to battery	▪ Rear climate control blower control circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear climate control blower control circuit for short circuit to power
B10BA-13	Rear Blower Control - Circuit open	▪ Rear climate control blower control circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear climate control blower control circuit for open circuit, high resistance
B11D1-4A	LIN Bus "C" - Incorrect component installed	▪ Incorrect component connected to the LIN bus	▪ Using the manufacturer approved diagnostic system, check for other LIN related DTCs and perform the relevant corrective action
B11D1-88	LIN Bus "C" - Bus off	▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B129A-4A	LIN Bus "D" - Incorrect component installed	▪ Incorrect component connected to the LIN bus	▪ Using the manufacturer approved diagnostic system, check for other LIN related DTCs and perform the relevant corrective action
B129A-88	LIN Bus "D" - Bus off	▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance

B12A6-11	Auxiliary Heater Water Pump - Circuit short to ground	▪ Fuel fired booster heater coolant pump circuit short circuit to ground	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault ▪ Refer to the electrical circuit diagrams and check the fuel fired booster heater coolant pump circuit for short circuit to ground ▪ Repair the circuit or install a new coolant pump located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B12A6-12	Auxiliary Heater Water Pump - Circuit short to battery	▪ Fuel fired booster heater coolant pump circuit short circuit to power	

			NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault ▪ Refer to the electrical circuit diagrams and check the fuel fired booster heater coolant pump circuit for short circuit to power ▪ Repair the circuit or install a new coolant pump located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B12A6-13	Auxiliary Heater Water Pump - Circuit open	▪ Fuel fired booster heater coolant pump circuit open circuit, high resistance	

			NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault ▪ Refer to the electrical circuit diagrams and check the fuel fired booster heater coolant pump circuit for open circuit, high resistance ▪ Repair the circuit or install a new coolant pump located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B12B1-11	Heated Washer Jet and Wiper Park Relay - Circuit short to ground	▪ Heated washer jet relay circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the heated washer jet relay circuit for short circuit to ground

B12B1-12	Heated Washer Jet and Wiper Park Relay - Circuit short to battery	Heated washer jet relay circuit short circuit to power	Refer to the electrical circuit diagrams and check the heated washer jet relay circuit for short circuit to power
B12B1-13	Heated Washer Jet and Wiper Park Relay - Circuit open	Heated washer jet relay circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the heated washer jet relay circuit for open circuit, high resistance
B12B2-11	AUX Heater Coolant Diverter Valve - Circuit short to ground	Fuel fired booster heater engine coolant bypass valve circuit short circuit to ground	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault - Refer to the electrical circuit diagrams and check the fuel fired booster heater engine coolant bypass valve circuit for short circuit to ground - Repair the circuit or install a new engine coolant bypass valve located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan.

			Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B12B2-12	AUX Heater Coolant Diverter Valve - Circuit short to battery	Fuel fired booster heater engine coolant bypass valve circuit short circuit to power	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault - Refer to the electrical circuit diagrams and check the fuel fired booster heater engine coolant bypass valve circuit for short circuit to power - Repair the circuit or install a new engine coolant bypass valve located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater

B12B2-13	AUX Heater Coolant Diverter Valve - Circuit open	Fuel fired booster heater engine coolant bypass valve circuit open circuit, high resistance	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault - Refer to the electrical circuit diagrams and check the fuel fired booster heater engine coolant bypass valve circuit for open circuit, high resistance - Repair the circuit or install a new engine coolant bypass valve located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B12B9-16	Auxiliary Heater Control Module - Circuit voltage below	Fuel fired booster heater power or	

threshold

ground circuit
open circuit,
high resistance

- Battery/charging system fault
- Fuel fired booster heater control module internal failure

NOTE:

The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault

- Refer to the electrical circuit diagrams and check the fuel fired booster heater power and ground circuits for open circuit, high resistance
- Refer to the relevant section of the workshop manual and test the battery and charging system
- Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
- If the fault persists, install a new fuel fired booster heater control module located within the fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM)

			Activation Test (0x900A)
B12B9-17	Auxiliary Heater Control Module - Circuit voltage above threshold	▪ Battery/charging system fault ▪ Fuel fired booster heater control module internal failure	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault ▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater ▪ If the fault persists, install a new fuel fired booster heater control module located within the fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)

B12B9-49	Auxiliary Heater Control Module - Internal electronic failure	Fuel fired booster heater control module internal failure	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater - If the fault persists, install a new fuel fired booster heater control module located within the fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
B12B9-64	Auxiliary Heater Control Module - Signal plausibility failure	Fuel fired booster heater overheated	

			NOTE: Consider whether environmental conditions may have caused this DTC to log before condemning the unit - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater - If the fault persists, install a new fuel fired booster heater control module located within the fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
B12B9-87	Auxiliary Heater Control Module - Missing message	- Fuel fired booster heater power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short	

circuit to power,
open circuit,
high resistance

- Fuel fired
booster heater
control module
internal failure

NOTE:

The fuel fired booster
heater assembly
contains many
serviceable
components. It should
NOT be necessary to
replace the entire fuel
fired booster heater
unit to rectify this fault

- Refer to the electrical circuit
diagrams and check the fuel
fired booster heater power
and ground circuits for open
circuit, high resistance
- Refer to the electrical circuit
diagrams and check the LIN
circuit for short circuit to
ground, short circuit to
power, open circuit, high
resistance
- Using the manufacturer
approved diagnostic system,
clear the DTCs and perform
routine - Auxiliary Heater
(AHCM) Activation Test
(0x900A). This will operate
the fuel fired booster heater
for 15 minutes. Note that
the routine should be
started only when the
engine is cool. The routine
should be performed with
the engine running and with
the climate controls set to
maximum heat and
maximum fan. Repeat the
routine again if smoke
continues to be visible from
the fuel fired booster heater
exhaust pipe or from the
area surrounding the fuel
fired booster heater
- If the fault persists, install a
new fuel fired booster
heater control module
located within the fuel fired
booster heater, clear the

			DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
B1352-49	Left Front Temperature Damper Motor - Internal electronic failure	Front left temperature blend motor internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left temperature blend motor
B1352-87	Left Front Temperature Damper Motor - Missing message	- Front left temperature blend motor power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front left temperature blend motor internal failure	- Refer to the electrical circuit diagrams and check the front left temperature blend motor power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left temperature blend motor
B1352-97	Left Front Temperature Damper Motor - Component or system operation obstructed or blocked	Front left temperature blend motor jammed or broken	Remove any obstructions from the front left temperature blend motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Left Front Air Temperature Blend Flap Actuator Requested Position (0x9864) - and - Left Front Air Temperature Blend Flap Actuator Measured Position (0x986F). If inconsistent, install a new front left temperature blend motor

B1354-49	Right Rear Temperature Damper Motor - Internal electronic failure	▪ Rear right temperature blend motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right temperature blend motor
B1354-87	Right Rear Temperature Damper Motor - Missing message	▪ Rear right temperature blend motor power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear right temperature blend motor internal failure	▪ Refer to the electrical circuit diagrams and check the rear right temperature blend motor power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right temperature blend motor
B1354-97	Right Rear Temperature Damper Motor - Component or system operation obstructed or blocked	▪ Rear right temperature blend motor jammed or broken	▪ Remove any obstructions from the rear right temperature blend motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Right Rear Air Temperature Blend Flap Actuator Requested Position (0x9868) - and - Right Rear Air Temperature Blend Flap Actuator Measured Position (0x9867). If inconsistent, install a new rear right temperature blend motor
B1364-49	Left Rear Temperature Damper Motor - Internal electronic failure	▪ Rear left temperature blend motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left temperature

			blend motor
B1364-87	Left Rear Temperature Damper Motor - Missing message	▪ Rear left temperature blend motor power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear left temperature blend motor internal failure	▪ Refer to the electrical circuit diagrams and check the rear left temperature blend motor power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left temperature blend motor
B1364-97	Left Rear Temperature Damper Motor - Component or system operation obstructed or blocked	▪ Rear left temperature blend motor jammed or broken	▪ Remove any obstructions from the rear left temperature blend motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Left Rear Air Temperature Blend Flap Actuator Requested Position (0x9866) - and - Left Rear Air Temperature Blend Flap Actuator Measured Position (0x9865). If inconsistent, install a new rear left temperature blend motor
B1366-49	Right Front Temperature Damper Motor - Internal electronic failure	▪ Front right temperature blend motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right temperature blend motor
B1366-87	Right Front Temperature Damper	▪ Front right temperature	▪ Refer to the electrical circuit diagrams and check the

	Motor - Missing message	blend motor power or ground circuit open circuit, high resistance ■ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Front right temperature blend motor internal failure	front right temperature blend motor power and ground circuits for open circuit, high resistance ■ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right temperature blend motor
B1366-97	Right Front Temperature Damper Motor - Component or system operation obstructed or blocked	■ Front right temperature blend motor jammed or broken	■ Remove any obstructions from the front right temperature blend motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Right Front Air Temperature Blend Flap Actuator Requested Position (0x9863) - and - Right Front Air Temperature Blend Flap Actuator Measured Position (0x986E). If inconsistent, install a new front right temperature blend motor
B13C2-49	Front Windscreen/Windshield Misting Sensor - Internal electronic failure	■ Humidity sensor internal failure	■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new humidity sensor
B13C2-87	Front Windscreen/Windshield Misting Sensor - Missing message	■ Humidity sensor power or ground circuit open circuit, high resistance ■ LIN circuit short circuit to	■ Refer to the electrical circuit diagrams and check the humidity sensor power and ground circuits for open circuit, high resistance ■ Refer to the electrical circuit diagrams and check the LIN

		ground, short circuit to power, open circuit, high resistance ▪ Humidity sensor internal failure	circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new humidity sensor
B13E2-11	Left Front Face Duct Air Temperature Sensor - Circuit short to ground	▪ Front left face duct air temperature sensor signal circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the front left face duct air temperature sensor signal circuit for short circuit to ground
B13E2-15	Left Front Face Duct Air Temperature Sensor - Circuit short to battery or open	▪ Front left face duct air temperature sensor signal circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the front left face duct air temperature sensor signal circuit for short circuit to power, open circuit, high resistance
B13E3-11	Right Front Face Duct Air Temperature Sensor - Circuit short to ground	▪ Front right face duct air temperature sensor signal circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the front right face duct air temperature sensor signal circuit for short circuit to ground
B13E3-15	Right Front Face Duct Air Temperature Sensor - Circuit short to battery or open	▪ Front right face duct air temperature sensor signal circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the front right face duct air temperature sensor signal circuit for short circuit to power, open circuit, high resistance
B13E8-11	Left Front Foot Duct Air Temperature Sensor - Circuit short to ground	▪ Front left foot duct air temperature sensor signal circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the front left foot duct air temperature sensor signal circuit for short circuit to ground

B13E8-15	Left Front Foot Duct Air Temperature Sensor - Circuit short to battery or open	Front left foot duct air temperature sensor signal circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front left foot duct air temperature sensor signal circuit for short circuit to power, open circuit, high resistance
B13E9-11	Right Front Foot Duct Air Temperature Sensor - Circuit short to ground	Front right foot duct air temperature sensor signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front right foot duct air temperature sensor signal circuit for short circuit to ground
B13E9-15	Right Front Foot Duct Air Temperature Sensor - Circuit short to battery or open	Front right foot duct air temperature sensor signal circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right foot duct air temperature sensor signal circuit for short circuit to power, open circuit, high resistance
B13EA-11	Left Rear Face Duct Air Temperature Sensor - Circuit short to ground	Rear left face duct air temperature sensor signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the rear left face duct air temperature sensor signal circuit for short circuit to ground
B13EA-15	Left Rear Face Duct Air Temperature Sensor - Circuit short to battery or open	Rear left face duct air temperature sensor signal circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear left face duct air temperature sensor signal circuit for short circuit to power, open circuit, high resistance
B13EB-11	Right Rear Face Duct Air Temperature Sensor - Circuit short to ground	Rear right face duct air temperature sensor signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the rear right face duct air temperature sensor signal circuit for short circuit to ground

B13EB-15	Right Rear Face Duct Air Temperature Sensor - Circuit short to battery or open	▪ Rear right face duct air temperature sensor signal circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right face duct air temperature sensor signal circuit for short circuit to power, open circuit, high resistance
B13EC-11	Left Rear Foot Duct Air Temperature Sensor - Circuit short to ground	▪ Rear left foot duct air temperature sensor signal circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear left foot duct air temperature sensor signal circuit for short circuit to ground
B13EC-15	Left Rear Foot Duct Air Temperature Sensor - Circuit short to battery or open	▪ Rear left foot duct air temperature sensor signal circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear left foot duct air temperature sensor signal circuit for short circuit to power, open circuit, high resistance
B13ED-11	Right Rear Foot Duct Air Temperature Sensor - Circuit short to ground	▪ Rear right foot duct air temperature sensor signal circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear right foot duct air temperature sensor signal circuit for short circuit to ground
B13ED-15	Right Rear Foot Duct Air Temperature Sensor - Circuit short to battery or open	▪ Rear right foot duct air temperature sensor signal circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right foot duct air temperature sensor signal circuit for short circuit to power, open circuit, high resistance
B13EE-49	Front Cold Air Bypass Flap Actuator Motor - Internal electronic failure	▪ Cool air bypass motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new cool air bypass motor

B13EE-87	Front Cold Air Bypass Flap Actuator Motor - Missing message	▪ Cool air bypass motor power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Cool air bypass motor internal failure	▪ Refer to the electrical circuit diagrams and check the cool air bypass motor power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new cool air bypass motor
B13EE-97	Front Cold Air Bypass Flap Actuator Motor - Component or system operation obstructed or blocked	▪ Cool air bypass motor jammed or broken	▪ Remove any obstructions from the cool air bypass motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Front Cold Air Bypass Flap Actuator Requested Position (0x985C) - and - Front Cold Air Bypass Flap Actuator Measured Position (0x985B). If inconsistent, install a new cool air bypass motor
B13EF-49	Left Front Distribution Flap Actuator Motor - Internal electronic failure	▪ Front left face/feet distribution motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left face/feet distribution motor
B13EF-87	Left Front Distribution Flap Actuator Motor - Missing message	▪ Front left face/feet distribution motor power or ground circuit open circuit, high resistance ▪ LIN circuit short	▪ Refer to the electrical circuit diagrams and check the front left face/feet distribution motor power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN

		circuit to ground, short circuit to power, open circuit, high resistance Front left face/feet distribution motor internal failure	circuit for short circuit to ground, short circuit to power, open circuit, high resistance Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left face/feet distribution motor
B13EF-97	Left Front Distribution Flap Actuator Motor - Component or system operation obstructed or blocked	Front left face/feet distribution motor jammed or broken	Remove any obstructions from the front left face/feet distribution motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Left Front Distribution Air Flap Actuator Requested Position (0x985E) - and - Left Front Distribution Air Flap Actuator Measured Position (0x985D). If inconsistent, install a new front left face/feet distribution motor
B13F0-49	Right Front Distribution Flap Actuator Motor - Internal electronic failure	Front right face/feet distribution motor internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right face/feet distribution motor
B13F0-87	Right Front Distribution Flap Actuator Motor - Missing message	- Front right face/feet distribution motor power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and check the front right face/feet distribution motor power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer

		Front right face/feet distribution motor internal failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right face/feet distribution motor
B13F0-97	Right Front Distribution Flap Actuator Motor - Component or system operation obstructed or blocked	Front right face/feet distribution motor jammed or broken	Remove any obstructions from the front right face/feet distribution motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Right Front Distribution Air Flap Actuator Requested Position (0x9860) - and - Right Front Distribution Air Flap Actuator Measured Position (0x985F). If inconsistent, install a new front right face/feet distribution motor
B13F1-49	Auxiliary Distribution Flap Actuator Motor - Internal electronic failure	Auxiliary climate control face/feet distribution motor internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new auxiliary climate control face/feet distribution motor
B13F1-87	Auxiliary Distribution Flap Actuator Motor - Missing message	- Auxiliary climate control face/feet distribution motor power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance - Auxiliary climate control face/feet distribution motor internal failure	- Refer to the electrical circuit diagrams and check the auxiliary climate control face/feet distribution motor power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new auxiliary climate control face/feet distribution motor

B13F1-97	Auxiliary Distribution Flap Actuator Motor - Component or system operation obstructed or blocked	▪ Auxiliary climate control face/feet distribution motor jammed or broken	▪ Remove any obstructions from the auxiliary climate control face/feet distribution motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Auxiliary Distribution Air Flap Actuator Requested Position (0x986D) - and - Auxiliary Distribution Air Flap Actuator Measured Position (0x9872). If inconsistent, install a new auxiliary climate control face/feet distribution motor
B13F2-49	Left Auxiliary Air Temperature Blend Flap Actuator Motor - Internal electronic failure	▪ Auxiliary climate control left temperature blend motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new auxiliary climate control left temperature blend motor
B13F2-87	Left Auxiliary Air Temperature Blend Flap Actuator Motor - Missing message	▪ Auxiliary climate control left temperature blend motor power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Auxiliary climate control left temperature blend motor internal failure	▪ Refer to the electrical circuit diagrams and check the auxiliary climate control left temperature blend motor power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new auxiliary climate control left temperature blend motor

B13F2-97	Left Auxiliary Air Temperature Blend Flap Actuator Motor - Component or system operation obstructed or blocked	▪ Auxiliary climate control left temperature blend motor jammed or broken	▪ Remove any obstructions from the auxiliary climate control left temperature blend motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Left Auxiliary Air Temperature Blend Flap Actuator Requested Position (0x986B) - and - Left Auxiliary Air Temperature Blend Flap Actuator Measured Position (0x9870). If inconsistent, install a new auxiliary climate control left temperature blend motor
B13F3-49	Right Auxiliary Air Temperature Blend Flap Actuator Motor - Internal electronic failure	▪ Auxiliary climate control right temperature blend motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new auxiliary climate control right temperature blend motor
B13F3-87	Right Auxiliary Air Temperature Blend Flap Actuator Motor - Missing message	▪ Auxiliary climate control right temperature blend motor power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Auxiliary climate control right temperature blend motor internal failure	▪ Refer to the electrical circuit diagrams and check the auxiliary climate control right temperature blend motor power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new auxiliary climate control right temperature blend motor
B13F3-	Right Auxiliary Air	▪ Auxiliary climate	▪ Remove any obstructions

97	Temperature Blend Flap Actuator Motor - Component or system operation obstructed or blocked	control right temperature blend motor jammed or broken	from the auxiliary climate control right temperature blend motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Right Auxiliary Air Temperature Blend Flap Actuator Requested Position (0x986C) - and - Right Auxiliary Air Temperature Blend Flap Actuator Measured Position (0x9871). If inconsistent, install a new auxiliary climate control right temperature blend motor
B13F4-11	Electric (PTC) Heater Control Circuit #1 - Circuit short to ground	Electric booster heater relay 1 circuit short circuit to ground	Refer to the electrical circuit diagrams and check the electric booster heater relay 1 circuit for short circuit to ground
B13F4-12	Electric (PTC) Heater Control Circuit #1 - Circuit short to battery	Electric booster heater relay 1 circuit short circuit to power	Refer to the electrical circuit diagrams and check the electric booster heater relay 1 circuit for short circuit to power
B13F4-13	Electric (PTC) Heater Control Circuit #1 - Circuit open	Electric booster heater relay 1 circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the electric booster heater relay 1 circuit for open circuit, high resistance
B13F5-11	Electric (PTC) Heater Control Circuit #2 - Circuit short to ground	Electric booster heater relay 2 circuit short circuit to ground	Refer to the electrical circuit diagrams and check the electric booster heater relay 2 circuit for short circuit to ground
B13F5-12	Electric (PTC) Heater Control Circuit #2 - Circuit short to battery	Electric booster heater relay 2 circuit short circuit to power	Refer to the electrical circuit diagrams and check the electric booster heater relay 2 circuit for short circuit to power

B13F5-13	Electric (PTC) Heater Control Circuit #2 - Circuit open	▪ Electric booster heater relay 2 circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the electric booster heater relay 2 circuit for open circuit, high resistance
B13FF-12	Left Rear Solar Sensor - Circuit short to battery	▪ Rear left sunload sensor circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear left sunload sensor circuit for short circuit to power
B13FF-14	Left Rear Solar Sensor - Circuit short to ground or open	▪ Rear left sunload sensor circuit short circuit to ground, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear left sunload sensor circuit for short circuit to ground, open circuit, high resistance
B1400-12	Right Rear Solar Sensor - Circuit short to battery	▪ Rear right sunload sensor circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear right sunload sensor circuit for short circuit to power
B1400-14	Right Rear Solar Sensor - Circuit short to ground or open	▪ Rear right sunload sensor circuit short circuit to ground, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right sunload sensor circuit for short circuit to ground, open circuit, high resistance
B1406-88	Park Heater Remote Receiver Module - Bus off	▪ Fuel fired booster heater receiver power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel fired booster heater	▪ Refer to the electrical circuit diagrams and check the fuel fired booster heater receiver power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system,

		receiver internal failure	clear the DTCs and retest. If the fault persists, install a new fuel fired booster heater receiver
B1406-96	Park Heater Remote Receiver Module - Component internal failure	Fuel fired booster heater receiver internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new fuel fired booster heater receiver
B140F-08	Electric A/C Compressor - Bus signal / message failures	- Electric air conditioning compressor power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance - Electric air conditioning compressor internal failure	- Refer to the electrical circuit diagrams and check the electric air conditioning compressor power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric air conditioning compressor
B140F-49	Electric A/C Compressor - Internal electronic failure	Electric air conditioning compressor internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric air conditioning compressor
B140F-87	Electric A/C Compressor - Missing message	- Electric air conditioning compressor power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short	- Refer to the electrical circuit diagrams and check the electric air conditioning compressor power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to

		circuit to power, open circuit, high resistance ▪ Electric air conditioning compressor internal failure	power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric air conditioning compressor
B1410-11	Front Refrigerant Magnetic Valve - Circuit short to ground	▪ Climate control isolation valve circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the climate control isolation valve circuit for short circuit to ground
B1410-12	Front Refrigerant Magnetic Valve - Circuit short to battery	▪ Climate control isolation valve circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the climate control isolation valve circuit for short circuit to power
B1410-13	Front Refrigerant Magnetic Valve - Circuit open	▪ Climate control isolation valve circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the climate control isolation valve circuit for open circuit, high resistance
B1411-11	Battery Chiller Magnetic Valve - Circuit short to ground	▪ Hybrid electric vehicle battery isolation valve circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the Hybrid electric vehicle battery isolation valve circuit for short circuit to ground
B1411-12	Battery Chiller Magnetic Valve - Circuit short to battery	▪ Hybrid electric vehicle battery isolation valve circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the Hybrid electric vehicle battery isolation valve circuit for short circuit to power
B1411-13	Battery Chiller Magnetic Valve - Circuit open	▪ Hybrid electric vehicle battery isolation valve circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the Hybrid electric vehicle battery isolation valve circuit for open circuit, high resistance

B1414-11	Left Front Cooled/Heated Seat Module Blower #2 Motor Speed Control - Circuit short to ground	Front left seat cushion climate assembly blower control circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front left seat cushion climate assembly blower control circuit for short circuit to ground
B1414-15	Left Front Cooled/Heated Seat Module Blower #2 Motor Speed Control - Circuit short to battery or open	Front left seat cushion climate assembly blower control circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front left seat cushion climate assembly blower control circuit for short circuit to power, open circuit, high resistance
B1415-49	Left Front Cooled/Heated Seat Control Module - Internal electronic failure	Front left seat climate control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left seat climate control module
B1415-55	Left Front Cooled/Heated Seat Control Module - Not configured	Front left seat climate control module is not programmed	Using the manufacturer approved diagnostic system, re-configure the front left seat climate control module with the latest level software
B1415-87	Left Front Cooled/Heated Seat Control Module - Missing message	- Front left seat climate control module power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front left seat climate control module internal failure	- Refer to the electrical circuit diagrams and check the front left seat climate control module power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left seat climate control module

B1415-93	Left Front Cooled/Heated Seat Control Module - No operation	- Front left seat climate control module power or ground circuit open circuit, high resistance - Front left seat climate control module internal failure	- Refer to the electrical circuit diagrams and check the front left seat climate control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left seat climate control module
B1416-11	Left Front Cooled/Heated Seat Thermal Electric Device #2 Temperature Sensor - Circuit short to ground	Front left seat cushion climate assembly temperature sensor circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front left seat cushion climate assembly temperature sensor circuit for short circuit to ground
B1416-15	Left Front Cooled/Heated Seat Thermal Electric Device #2 Temperature Sensor - Circuit short to battery or open	Front left seat cushion climate assembly temperature sensor circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front left seat cushion climate assembly temperature sensor circuit for short circuit to power, open circuit, high resistance
B1417-11	Left Front Cooled/Heated Seat Thermal Electric Device #1 Temperature Sensor - Circuit short to ground	Front left seat backrest climate assembly temperature sensor circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front left seat backrest climate assembly temperature sensor circuit for short circuit to ground
B1417-15	Left Front Cooled/Heated Seat Thermal Electric Device #1 Temperature Sensor - Circuit short to battery or open	Front left seat backrest climate assembly temperature sensor circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front left seat backrest climate assembly temperature sensor circuit for short circuit to power, open circuit, high resistance

B1418-49	Right Front Cooled/Heated Seat Control Module - Internal electronic failure	Front right seat climate control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right seat climate control module
B1418-55	Right Front Cooled/Heated Seat Control Module - Not configured	Front right seat climate control module is not programmed	Using the manufacturer approved diagnostic system, re-configure the front right seat climate control module with the latest level software
B1418-87	Right Front Cooled/Heated Seat Control Module - Missing message	- Front right seat climate control module power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front right seat climate control module internal failure	- Refer to the electrical circuit diagrams and check the front right seat climate control module power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right seat climate control module
B1418-93	Right Front Cooled/Heated Seat Control Module - No operation	- Front right seat climate control module power or ground circuit open circuit, high resistance - Front right seat climate control module internal failure	- Refer to the electrical circuit diagrams and check the front right seat climate control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right seat climate control module
B1419-49	Left Rear Cooled/Heated Seat	Rear left seat climate control	Using the manufacturer approved diagnostic system,

	Control Module - Internal electronic failure	module internal failure	clear the DTCs and retest. If the fault persists, install a new rear left seat climate control module
B1419-55	Left Rear Cooled/Heated Seat Control Module - Not configured	■ Rear left seat climate control module is not programmed	■ Using the manufacturer approved diagnostic system, re-configure the rear left seat climate control module with the latest level software
B1419-87	Left Rear Cooled/Heated Seat Control Module - Missing message	■ Rear left seat climate control module power or ground circuit open circuit, high resistance ■ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Rear left seat climate control module internal failure	■ Refer to the electrical circuit diagrams and check the rear left seat climate control module power and ground circuits for open circuit, high resistance ■ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left seat climate control module
B1419-93	Left Rear Cooled/Heated Seat Control Module - No operation	■ Rear left seat climate control module power or ground circuit open circuit, high resistance ■ Rear left seat climate control module internal failure	■ Refer to the electrical circuit diagrams and check the rear left seat climate control module power and ground circuits for open circuit, high resistance ■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left seat climate control module
B141A-49	Right Rear Cooled/Heated Seat Control Module - Internal electronic failure	■ Rear right seat climate control module internal failure	■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right seat climate

			control module
B141A-55	Right Rear Cooled/Heated Seat Control Module - Not configured	▪ Rear right seat climate control module is not programmed	▪ Using the manufacturer approved diagnostic system, re-configure the rear right seat climate control module with the latest level software
B141A-87	Right Rear Cooled/Heated Seat Control Module - Missing message	▪ Rear right seat climate control module power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear right seat climate control module internal failure	▪ Refer to the electrical circuit diagrams and check the rear right seat climate control module power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right seat climate control module
B141A-93	Right Rear Cooled/Heated Seat Control Module - No operation	▪ Rear right seat climate control module power or ground circuit open circuit, high resistance ▪ Rear right seat climate control module internal failure	▪ Refer to the electrical circuit diagrams and check the rear right seat climate control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right seat climate control module
B141B-11	Right Front Cooled/Heated Seat Thermal Electric Device #2 Temperature Sensor - Circuit short to ground	▪ Front right seat cushion climate assembly temperature sensor circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the front right seat cushion climate assembly temperature sensor circuit for short circuit to ground

B141B-15	Right Front Cooled/Heated Seat Thermal Electric Device #2 Temperature Sensor - Circuit short to battery or open	Front right seat cushion climate assembly temperature sensor circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right seat cushion climate assembly temperature sensor circuit for short circuit to power, open circuit, high resistance
B141C-11	Right Front Cooled/Heated Seat Thermal Electric Device #1 Temperature Sensor - Circuit short to ground	Front right seat backrest climate assembly temperature sensor circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front right seat backrest climate assembly temperature sensor circuit for short circuit to ground
B141C-15	Right Front Cooled/Heated Seat Thermal Electric Device #1 Temperature Sensor - Circuit short to battery or open	Front right seat backrest climate assembly temperature sensor circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right seat backrest climate assembly temperature sensor circuit for short circuit to power, open circuit, high resistance
B141D-11	Left Rear Cooled/Heated Seat Thermal Electric Device #2 Temperature Sensor - Circuit short to ground	Rear left seat cushion climate assembly temperature sensor circuit short circuit to ground	Refer to the electrical circuit diagrams and check the rear left seat cushion climate assembly temperature sensor circuit for short circuit to ground
B141D-15	Left Rear Cooled/Heated Seat Thermal Electric Device #2 Temperature Sensor - Circuit short to battery or open	Rear left seat cushion climate assembly temperature sensor circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear left seat cushion climate assembly temperature sensor circuit for short circuit to power, open circuit, high resistance
B141E-11	Left Rear Cooled/Heated Seat	Rear left seat backrest climate	Refer to the electrical circuit diagrams and check the rear

	Thermal Electric Device #1 Temperature Sensor - Circuit short to ground	assembly temperature sensor circuit short circuit to ground	left seat backrest climate assembly temperature sensor circuit for short circuit to ground
B141E-15	Left Rear Cooled/Heated Seat Thermal Electric Device #1 Temperature Sensor - Circuit short to battery or open	▪ Rear left seat backrest climate assembly temperature sensor circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear left seat backrest climate assembly temperature sensor circuit for short circuit to power, open circuit, high resistance
B141F-11	Right Rear Cooled/Heated Seat Thermal Electric Device #2 Temperature Sensor - Circuit short to ground	▪ Rear right seat cushion climate assembly temperature sensor circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear right seat cushion climate assembly temperature sensor circuit for short circuit to ground
B141F-15	Right Rear Cooled/Heated Seat Thermal Electric Device #2 Temperature Sensor - Circuit short to battery or open	▪ Rear right seat cushion climate assembly temperature sensor circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right seat cushion climate assembly temperature sensor circuit for short circuit to power, open circuit, high resistance
B1420-11	Right Rear Cooled/Heated Seat Thermal Electric Device #1 Temperature Sensor - Circuit short to ground	▪ Rear right seat backrest climate assembly temperature sensor circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear right seat backrest climate assembly temperature sensor circuit for short circuit to ground
B1420-15	Right Rear Cooled/Heated Seat Thermal Electric Device #1 Temperature Sensor - Circuit short to battery or open	▪ Rear right seat backrest climate assembly temperature sensor circuit short circuit to power, open	▪ Refer to the electrical circuit diagrams and check the rear right seat backrest climate assembly temperature sensor circuit for short circuit to power, open circuit, high resistance

		circuit, high resistance	
B1421-11	Left Front Cooled/Heated Seat Thermal Electric Device #2 - Circuit short to ground	Front left seat cushion climate assembly thermal electric device circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front left seat cushion climate assembly thermal electric device circuit for short circuit to ground
B1421-12	Left Front Cooled/Heated Seat Thermal Electric Device #2 - Circuit short to battery	Front left seat cushion climate assembly thermal electric device circuit short circuit to power	Refer to the electrical circuit diagrams and check the front left seat cushion climate assembly thermal electric device circuit for short circuit to power
B1421-13	Left Front Cooled/Heated Seat Thermal Electric Device #2 - Circuit open	Front left seat cushion climate assembly thermal electric device circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the front left seat cushion climate assembly thermal electric device circuit for open circuit, high resistance
B1421-4B	Left Front Cooled/Heated Seat Thermal Electric Device #2 - Over temperature	Front left seat cushion climate assembly thermal electric device internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left seat cushion climate assembly thermal electric device
B1422-11	Left Front Cooled/Heated Seat Thermal Electric Device #1 - Circuit short to ground	Front left seat backrest climate assembly thermal electric device circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front left seat backrest climate assembly thermal electric device circuit for short circuit to ground
B1422-12	Left Front Cooled/Heated Seat Thermal Electric Device #1 - Circuit short to battery	Front left seat backrest climate assembly thermal electric device circuit	Refer to the electrical circuit diagrams and check the front left seat backrest climate assembly thermal electric device circuit for

		short circuit to power	short circuit to power
B1422-13	Left Front Cooled/Heated Seat Thermal Electric Device #1 - Circuit open	Front left seat backrest climate assembly thermal electric device circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the front left seat backrest climate assembly thermal electric device circuit for open circuit, high resistance
B1422-4B	Left Front Cooled/Heated Seat Thermal Electric Device #1 - Over temperature	Front left seat backrest climate assembly thermal electric device internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left seat backrest climate assembly thermal electric device
B1423-11	Left Front Cooled/Heated Seat Module Blower #1 Motor Speed Control - Circuit short to ground	Front left seat backrest climate assembly blower control circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front left seat backrest climate assembly blower control circuit for short circuit to ground
B1423-15	Left Front Cooled/Heated Seat Module Blower #1 Motor Speed Control - Circuit short to battery or open	Front left seat backrest climate assembly blower control circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front left seat backrest climate assembly blower control circuit for short circuit to power, open circuit, high resistance
B1424-11	Left Front Cooled/Heated Seat Module Blower Motor Power - Circuit short to ground	Front left seat climate assembly blower circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front left seat climate assembly blower circuit for short circuit to ground
B1424-15	Left Front Cooled/Heated Seat Module Blower Motor Power - Circuit short to battery or open	Front left seat climate assembly blower circuit short circuit to power, open circuit,	Refer to the electrical circuit diagrams and check the front left seat climate assembly blower circuit for short circuit to power, open circuit, high resistance

		high resistance	
B1425-11	Right Front Cooled/Heated Seat Thermal Electric Device #2 - Circuit short to ground	Front right seat cushion climate assembly thermal electric device circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front right seat cushion climate assembly thermal electric device circuit for short circuit to ground
B1425-12	Right Front Cooled/Heated Seat Thermal Electric Device #2 - Circuit short to battery	Front right seat cushion climate assembly thermal electric device circuit short circuit to power	Refer to the electrical circuit diagrams and check the front right seat cushion climate assembly thermal electric device circuit for short circuit to power
B1425-13	Right Front Cooled/Heated Seat Thermal Electric Device #2 - Circuit open	Front right seat cushion climate assembly thermal electric device circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right seat cushion climate assembly thermal electric device circuit for open circuit, high resistance
B1425-4B	Right Front Cooled/Heated Seat Thermal Electric Device #2 - Over temperature	Front right seat cushion climate assembly thermal electric device internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right seat cushion climate assembly thermal electric device
B1426-11	Right Front Cooled/Heated Seat Thermal Electric Device #1 - Circuit short to ground	Front right seat backrest climate assembly thermal electric device circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front right seat backrest climate assembly thermal electric device circuit for short circuit to ground
B1426-12	Right Front Cooled/Heated Seat Thermal Electric Device #1 - Circuit short to battery	Front right seat backrest climate assembly thermal electric device circuit short circuit to	Refer to the electrical circuit diagrams and check the front right seat backrest climate assembly thermal electric device circuit for short circuit to power

		power	
B1426-13	Right Front Cooled/Heated Seat Thermal Electric Device #1 - Circuit open	Front right seat backrest climate assembly thermal electric device circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right seat backrest climate assembly thermal electric device circuit for open circuit, high resistance
B1426-4B	Right Front Cooled/Heated Seat Thermal Electric Device #1 - Over temperature	Front right seat backrest climate assembly thermal electric device internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right seat backrest climate assembly thermal electric device
B1427-11	Right Front Cooled/Heated Seat Module Blower Motor Power - Circuit short to ground	Front right seat climate assembly blower circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front right seat climate assembly blower circuit for short circuit to ground
B1427-15	Right Front Cooled/Heated Seat Module Blower Motor Power - Circuit short to battery or open	Front right seat climate assembly blower circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right seat climate assembly blower circuit for short circuit to power, open circuit, high resistance
B1428-11	Right Front Cooled/Heated Seat Module Blower #1 Motor Speed Control - Circuit short to ground	Front right seat backrest climate assembly blower control circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front right seat backrest climate assembly blower control circuit for short circuit to ground
B1428-15	Right Front Cooled/Heated Seat Module Blower #1 Motor Speed Control - Circuit short to battery or open	Front right seat backrest climate assembly blower control circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right seat backrest climate assembly blower control circuit for short circuit to power, open circuit, high resistance

B1429-11	Right Front Cooled/Heated Seat Module Blower #2 Motor Speed Control - Circuit short to ground	Front right seat cushion climate assembly blower control circuit short circuit to ground	Refer to the electrical circuit diagrams and check the front right seat cushion climate assembly blower control circuit for short circuit to ground
B1429-15	Right Front Cooled/Heated Seat Module Blower #2 Motor Speed Control - Circuit short to battery or open	Front right seat cushion climate assembly blower control circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right seat cushion climate assembly blower control circuit for short circuit to power, open circuit, high resistance
B142A-11	Left Rear Cooled/Heated Seat Module Blower Motor Power - Circuit short to ground	Rear left seat climate assembly blower circuit short circuit to ground	Refer to the electrical circuit diagrams and check the rear left seat climate assembly blower circuit for short circuit to ground
B142A-15	Left Rear Cooled/Heated Seat Module Blower Motor Power - Circuit short to battery or open	Rear left seat climate assembly blower circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear left seat climate assembly blower circuit for short circuit to power, open circuit, high resistance
B142B-11	Left Rear Cooled/Heated Seat Module Blower #1 Motor Speed Control - Circuit short to ground	Rear left seat backrest climate assembly blower control circuit short circuit to ground	Refer to the electrical circuit diagrams and check the rear left seat backrest climate assembly blower control circuit for short circuit to ground
B142B-15	Left Rear Cooled/Heated Seat Module Blower #1 Motor Speed Control - Circuit short to battery or open	Rear left seat backrest climate assembly blower control circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear left seat backrest climate assembly blower control circuit for short circuit to power, open circuit, high resistance

B142C-11	Left Rear Cooled/Heated Seat Module Blower #2 Motor Speed Control - Circuit short to ground	▪ Rear left seat cushion climate assembly blower control circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear left seat cushion climate assembly blower control circuit for short circuit to ground
B142C-15	Left Rear Cooled/Heated Seat Module Blower #2 Motor Speed Control - Circuit short to battery or open	▪ Rear left seat cushion climate assembly blower control circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear left seat cushion climate assembly blower control circuit for short circuit to power, open circuit, high resistance
B142D-11	Left Rear Cooled/Heated Seat Thermal Electric Device #1 - Circuit short to ground	▪ Rear left seat backrest climate assembly thermal electric device circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear left seat backrest climate assembly thermal electric device circuit for short circuit to ground
B142D-12	Left Rear Cooled/Heated Seat Thermal Electric Device #1 - Circuit short to battery	▪ Rear left seat backrest climate assembly thermal electric device circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear left seat backrest climate assembly thermal electric device circuit for short circuit to power
B142D-13	Left Rear Cooled/Heated Seat Thermal Electric Device #1 - Circuit open	▪ Rear left seat backrest climate assembly thermal electric device circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear left seat backrest climate assembly thermal electric device circuit for open circuit, high resistance
B142D-4B	Left Rear Cooled/Heated Seat Thermal Electric Device #1 - Over temperature	▪ Rear left seat backrest climate assembly thermal electric device internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left seat backrest climate assembly thermal electric device

B142E-11	Left Rear Cooled/Heated Seat Thermal Electric Device #2 - Circuit short to ground	▪ Rear left seat cushion climate assembly thermal electric device circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear left seat cushion climate assembly thermal electric device circuit for short circuit to ground
B142E-12	Left Rear Cooled/Heated Seat Thermal Electric Device #2 - Circuit short to battery	▪ Rear left seat cushion climate assembly thermal electric device circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear left seat cushion climate assembly thermal electric device circuit for short circuit to power
B142E-13	Left Rear Cooled/Heated Seat Thermal Electric Device #2 - Circuit open	▪ Rear left seat cushion climate assembly thermal electric device circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear left seat cushion climate assembly thermal electric device circuit for open circuit, high resistance
B142E-4B	Left Rear Cooled/Heated Seat Thermal Electric Device #2 - Over temperature	▪ Rear left seat cushion climate assembly thermal electric device internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left seat cushion climate assembly thermal electric device
B142F-11	Right Rear Cooled/Heated Seat Module Blower Motor Power - Circuit short to ground	▪ Rear right seat climate assembly blower circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear right seat climate assembly blower circuit for short circuit to ground
B142F-15	Right Rear Cooled/Heated Seat Module Blower Motor Power - Circuit short to battery or open	▪ Rear right seat climate assembly blower circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right seat climate assembly blower circuit for short circuit to power, open circuit, high resistance
B1430-11	Right Rear Cooled/Heated Seat	▪ Rear right seat backrest climate	▪ Refer to the electrical circuit diagrams and check the rear

	Module Blower #1 Motor Speed Control - Circuit short to ground	assembly blower control circuit short circuit to ground	right seat backrest climate assembly blower control circuit for short circuit to ground
B1430-15	Right Rear Cooled/Heated Seat Module Blower #1 Motor Speed Control - Circuit short to battery or open	▪ Rear right seat backrest climate assembly blower control circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right seat backrest climate assembly blower control circuit for short circuit to power, open circuit, high resistance
B1431-11	Right Rear Cooled/Heated Seat Module Blower #2 Motor Speed Control - Circuit short to ground	▪ Rear right seat cushion climate assembly blower control circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear right seat cushion climate assembly blower control circuit for short circuit to ground
B1431-15	Right Rear Cooled/Heated Seat Module Blower #2 Motor Speed Control - Circuit short to battery or open	▪ Rear right seat cushion climate assembly blower control circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right seat cushion climate assembly blower control circuit for short circuit to power, open circuit, high resistance
B1432-11	Right Rear Cooled/Heated Seat Thermal Electric Device #1 - Circuit short to ground	▪ Rear right seat backrest climate assembly thermal electric device circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear right seat backrest climate assembly thermal electric device circuit for short circuit to ground
B1432-12	Right Rear Cooled/Heated Seat Thermal Electric Device #1 - Circuit short to battery	▪ Rear right seat backrest climate assembly thermal electric device circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear right seat backrest climate assembly thermal electric device circuit for short circuit to power
B1432-13	Right Rear Cooled/Heated Seat	▪ Rear right seat backrest climate	▪ Refer to the electrical circuit diagrams and check the rear

	Thermal Electric Device #1 - Circuit open	assembly thermal electric device circuit open circuit, high resistance	right seat backrest climate assembly thermal electric device circuit for open circuit, high resistance
B1432- 4B	Right Rear Cooled/Heated Seat Thermal Electric Device #1 - Over temperature	▪ Rear right seat backrest climate assembly thermal electric device internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right seat backrest climate assembly thermal electric device
B1433- 11	Right Rear Cooled/Heated Seat Thermal Electric Device #2 - Circuit short to ground	▪ Rear right seat cushion climate assembly thermal electric device circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear right seat cushion climate assembly thermal electric device circuit for short circuit to ground
B1433- 12	Right Rear Cooled/Heated Seat Thermal Electric Device #2 - Circuit short to battery	▪ Rear right seat cushion climate assembly thermal electric device circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear right seat cushion climate assembly thermal electric device circuit for short circuit to power
B1433- 13	Right Rear Cooled/Heated Seat Thermal Electric Device #2 - Circuit open	▪ Rear right seat cushion climate assembly thermal electric device circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear right seat cushion climate assembly thermal electric device circuit for open circuit, high resistance
B1433- 4B	Right Rear Cooled/Heated Seat Thermal Electric Device #2 - Over temperature	▪ Rear right seat cushion climate assembly thermal electric device internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right seat cushion climate assembly thermal electric device
B1434- 49	Front Distribution Flap Actuator Motor -	▪ Front face/feet distribution	▪ Using the manufacturer approved diagnostic system,

	Internal electronic failure	motor internal failure	clear the DTCs and retest. If the fault persists, install a new front face/feet distribution motor
B1434-87	Front Distribution Flap Actuator Motor - Missing message	- Front face/feet distribution motor power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance - Front face/feet distribution motor internal failure	- Refer to the electrical circuit diagrams and check the front face/feet distribution motor power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front face/feet distribution motor
B1434-97	Front Distribution Flap Actuator Motor - Component or system operation obstructed or blocked	Front face/feet distribution motor jammed or broken	Remove any obstructions from the front face/feet distribution motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Front Distribution Flap Actuator Motor Requested Position (0x989B) - and - Front Distribution Flap Actuator Motor Measured Position (0x989C). If inconsistent, install a new front face/feet distribution motor
B1435-49	Rear Air Temperature Blend Flap Actuator Motor - Internal electronic failure	Rear temperature blend motor internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear temperature blend motor

B1435-87	Rear Air Temperature Blend Flap Actuator Motor - Missing message	▪ Rear temperature blend motor power or ground circuit open circuit, high resistance ▪ LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear temperature blend motor internal failure	▪ Refer to the electrical circuit diagrams and check the rear temperature blend motor power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear temperature blend motor
B1435-97	Rear Air Temperature Blend Flap Actuator Motor - Component or system operation obstructed or blocked	▪ Rear temperature blend motor jammed or broken	▪ Remove any obstructions from the rear temperature blend motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Rear Air Temperature Blend Flap Actuator Position Requested Position (0x989E) - and - Rear Air Temperature Blend Flap Actuator Position Measured Position (0x989D). If inconsistent, install a new rear temperature blend motor
B1A59-11	Sensor 5 Volt Supply - Circuit short to ground	▪ Refrigerant pressure sensor circuit short circuit to ground ▪ Front sunload sensor circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the refrigerant pressure sensor circuit for short circuit to ground ▪ Refer to the electrical circuit diagrams and check the front sunload sensor circuit for short circuit to ground
B1A59-12	Sensor 5 Volt Supply - Circuit short to battery	▪ Refrigerant pressure sensor	▪ Refer to the electrical circuit diagrams and check the

		circuit short circuit to power Front sunload sensor circuit short circuit to power	refrigerant pressure sensor circuit for short circuit to power Refer to the electrical circuit diagrams and check the front sunload sensor circuit for short circuit to power
B1A61-11	Cabin Temperature Sensor - Circuit short to ground	In-vehicle temperature sensor circuit short circuit to ground	Refer to the electrical circuit diagrams and check the in-vehicle temperature sensor circuit for short circuit to ground
B1A61-15	Cabin Temperature Sensor - Circuit short to battery or open	In-vehicle temperature sensor circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the in-vehicle temperature sensor circuit for short circuit to power, open circuit, high resistance
B1A63-12	Right Solar Sensor - Circuit short to battery	Front right sunload sensor circuit short circuit to power	Refer to the electrical circuit diagrams and check the front right sunload sensor circuit for short circuit to power
B1A63-14	Right Solar Sensor - Circuit short to ground or open	Front right sunload sensor circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the front right sunload sensor circuit for short circuit to ground, open circuit, high resistance
B1A64-12	Left Solar Sensor - Circuit short to battery	Front left sunload sensor circuit short circuit to power	Refer to the electrical circuit diagrams and check the front left sunload sensor circuit for short circuit to power
B1A64-14	Left Solar Sensor - Circuit short to ground or open	Front left sunload sensor circuit short circuit to ground, open	Refer to the electrical circuit diagrams and check the front left sunload sensor circuit for short circuit to ground, open circuit, high

		circuit, high resistance	resistance
B1B71-11	Evaporator Temperature Sensor - Circuit short to ground	Evaporator temperature sensor circuit short circuit to ground	Refer to the electrical circuit diagrams and check the evaporator temperature sensor circuit for short circuit to ground
B1B71-15	Evaporator Temperature Sensor - Circuit short to battery or open	Evaporator temperature sensor circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the evaporator temperature sensor circuit for short circuit to power, open circuit, high resistance
B1B72-11	LIN Bus #1 Power Supply Circuit - Circuit short to ground	LIN component supply circuit short circuit to ground	Refer to the electrical circuit diagrams and check the LIN component supply circuit for short circuit to ground
B1B72-12	LIN Bus #1 Power Supply Circuit - Circuit short to battery	LIN component supply circuit short circuit to power	Refer to the electrical circuit diagrams and check the LIN component supply circuit for short circuit to power
B1B7D-49	Rear Air Distribution Actuator - Internal electronic failure	Rear face/feet distribution motor internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear face/feet distribution motor
B1B7D-87	Rear Air Distribution Actuator - Missing message	- Rear face/feet distribution motor power or ground circuit open circuit, high resistance - LIN circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and check the rear face/feet distribution motor power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		▪ Rear face/feet distribution motor internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear face/feet distribution motor
B1B7D-97	Rear Air Distribution Actuator - Component or system operation obstructed or blocked	▪ Rear face/feet distribution motor jammed or broken	▪ Remove any obstructions from the rear face/feet distribution motor mechanism. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, compare datalogger signals - Rear Distribution Air Flap Actuator Requested Position (0x986A) - and - Rear Distribution Air Flap Actuator Measured Position (0x9869). If inconsistent, install a new rear face/feet distribution motor
B1C0B-11	Rear Refrigerant Magnetic Valve - Circuit short to ground	▪ Rear climate control magnetic valve circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear climate control magnetic valve circuit for short circuit to ground
B1C0B-12	Rear Refrigerant Magnetic Valve - Circuit short to battery	▪ Rear climate control magnetic valve circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear climate control magnetic valve circuit for short circuit to power
B1C0B-13	Rear Refrigerant Magnetic Valve - Circuit open	▪ Rear climate control magnetic valve circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear climate control magnetic valve circuit for open circuit, high resistance
B1D22-11	Coolant Temperature Sensor - Circuit short to ground	▪ Fuel fired booster heater coolant temperature sensor circuit	

		short circuit to ground	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault Refer to the electrical circuit diagrams and check the fuel fired booster heater coolant temperature sensor circuit for short circuit to ground. Repair the circuit or install a new coolant temperature sensor located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D22-15	Coolant Temperature Sensor - Circuit short to battery or open	Fuel fired booster heater coolant temperature sensor circuit short circuit to	

		power, open circuit, high resistance	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault Refer to the electrical circuit diagrams and check the fuel fired booster heater coolant temperature sensor circuit for short circuit to power, open circuit, high resistance. Repair the circuit or install a new coolant temperature sensor located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D23-11	Overheat Sensor - Circuit short to ground	Fuel fired booster heater overheat temperature sensor circuit	

		short circuit to ground	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault Refer to the electrical circuit diagrams and check the fuel fired booster heater overheat temperature sensor circuit for short circuit to ground. Repair the circuit or install a new overheat temperature sensor located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D23-15	Overheat Sensor - Circuit short to battery or open	Fuel fired booster heater overheat temperature sensor circuit short circuit to	

		power, open circuit, high resistance	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault Refer to the electrical circuit diagrams and check the fuel fired booster heater overheat temperature sensor circuit for short circuit to power, open circuit, high resistance. Repair the circuit or install a new overheat temperature sensor located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D23-85	Overheat Sensor - Signal above allowable range	Fuel fired booster heater internal failure	

			NOTE: Consider whether environmental conditions may have caused this DTC to log before condemning the unit Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new fuel fired booster heater. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D24-11	Glow Plug - Circuit short to ground	Fuel fired booster heater glow plug circuit short circuit to ground	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault

			Refer to the electrical circuit diagrams and check the fuel fired booster heater glow plug circuit for short circuit to ground. Repair the circuit or install a new glow plug located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D24-12	Glow Plug - Circuit short to battery	Fuel fired booster heater glow plug circuit short circuit to power	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault Refer to the electrical circuit diagrams and check the fuel fired booster heater glow plug circuit for short circuit to power. Repair the circuit or install a new glow plug located within the fuel fired

			booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D24-13	Glow Plug - Circuit open	Fuel fired booster heater glow plug circuit open circuit, high resistance	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault Refer to the electrical circuit diagrams and check the fuel fired booster heater glow plug circuit for open circuit, high resistance. Repair the circuit or install a new glow plug located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A).

			This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D24-64	Glow Plug - Signal plausibility failure	Fuel fired booster heater glow plug failure	NOTES: - The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault - This DTC is set when a flame is detected before combustion is activated - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the

			routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater. If the fault persists, install a new glow plug located within the fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A) and re-test ▪ If the fault persists, install a new fuel fired booster heater control module located within the fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A) and re-test
B1D25-11	Heater Fuel Pump - Circuit short to ground	▪ Fuel fired booster heater fuel pump circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the fuel fired booster heater fuel pump circuit for short circuit to ground. Repair the circuit or install a new fuel fired booster heater fuel pump as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater

B1D25-12	Heater Fuel Pump - Circuit short to battery	Fuel fired booster heater fuel pump circuit short circuit to power	Refer to the electrical circuit diagrams and check the fuel fired booster heater fuel pump circuit for short circuit to power. Repair the circuit or install a new fuel fired booster heater fuel pump as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D25-13	Heater Fuel Pump - Circuit open	Fuel fired booster heater fuel pump circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the fuel fired booster heater fuel pump circuit for open circuit, high resistance. Repair the circuit or install a new fuel fired booster heater fuel pump as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate

			controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater
B1D26-11	Combustion Air Blower - Circuit short to ground	■ Fuel fired booster heater blower contaminated with water or debris ingress following driving through deep water or extreme off-road driving ■ Fuel fired booster heater blower circuit short circuit to ground ■ Fuel fired booster heater evaporator assembly contaminated with water or debris ingress following driving through deep water or extreme off-road driving	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault ■ Check the fuel fired booster heater air intake and the fuel fired booster heater blower for signs of restriction from sediment or debris. Remove any foreign matter from the air intake and blower assembly. Clear DTC and retest ■ If fault persists, refer to the electrical circuit diagrams and check the fuel fired booster heater blower circuit for short circuit to ground. Repair the circuit or install a new blower located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be

			performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater ■ If the fuel fired booster heater does not operate correctly following replacement of the blower located within the fuel fired booster heater, replace the fuel fired booster heater combustion tube and evaporator assembly, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
B1D26-12	Combustion Air Blower - Circuit short to battery	■ Fuel fired booster heater blower contaminated with water or debris ingress following driving through deep water or extreme off-road driving ■ Fuel fired booster heater blower circuit short circuit to power ■ Fuel fired booster heater evaporator assembly contaminated with water or debris ingress following driving through deep water or extreme off-	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault ■ Check the fuel fired booster heater air intake and the fuel fired booster heater blower for signs of restriction from sediment or debris. Remove any foreign matter from the air intake and blower assembly. Clear DTC and retest ■ If fault persists, refer to the electrical circuit diagrams and check the fuel fired booster heater blower circuit

		road driving	for short circuit to power. Repair the circuit or install a new blower located within the fuel fired booster heater as necessary. Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater ▪ If the fuel fired booster heater does not operate correctly following replacement of the blower located within the fuel fired booster heater, replace the fuel fired booster heater combustion tube and evaporator assembly, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
B1D26-13	Combustion Air Blower - Circuit open	▪ Fuel fired booster heater blower contaminated with water or debris ingress following driving through deep water or extreme off-road driving ▪ Fuel fired	

booster heater
blower circuit
open circuit,
high resistance

- Fuel fired
booster heater
evaporator
assembly
contaminated
with water or
debris ingress
following driving
through deep
water or
extreme off-
road driving

NOTE:

The fuel fired booster
heater assembly
contains many
serviceable
components. It should
NOT be necessary to
replace the entire fuel
fired booster heater
unit to rectify this fault

- Check the fuel fired booster
heater air intake and the fuel
fired booster heater blower
for signs of restriction from
sediment or debris. Remove
any foreign matter from the
air intake and blower
assembly. Clear DTC and
retest
- If fault persists, refer to the
electrical circuit diagrams
and check the fuel fired
booster heater blower circuit
for open circuit, high
resistance. Install a new
blower located within the
fuel fired booster heater as
necessary. Using the
manufacturer approved
diagnostic system, clear the
DTCs and perform routine -
Auxiliary Heater (AHCM)
Activation Test (0x900A).
This will operate the fuel
fired booster heater for 15
minutes. Note that the
routine should be started
only when the engine is
cool. The routine should be
performed with the engine
running and with the climate
controls set to maximum
heat and maximum fan.
Repeat the routine again if
smoke continues to be
visible from the fuel fired
booster heater exhaust pipe
or from the area surrounding
the fuel fired booster heater

			▪ If the fuel fired booster heater does not operate correctly following replacement of the blower located within the fuel fired booster heater, replace the fuel fired booster heater combustion tube and evaporator assembly, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
B1D26-71	Combustion Air Blower - Actuator stuck	▪ Fuel fired booster heater blower contaminated with water or debris ingress following driving through deep water or extreme off-road driving ▪ Fuel fired booster heater blower jammed/seized ▪ Fuel fired booster heater evaporator assembly contaminated with water or debris ingress following driving through deep water or extreme off-road driving	NOTE: The fuel fired booster heater assembly contains many serviceable components. It should NOT be necessary to replace the entire fuel fired booster heater unit to rectify this fault ▪ Check the fuel fired booster heater air intake and the fuel fired booster heater blower for signs of restriction from sediment or debris. Remove any foreign matter from the air intake and blower assembly ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the

			routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater. If the fault persists, install a new blower located within the fuel fired booster heater as necessary ▪ If the fuel fired booster heater does not operate correctly following replacement of the blower located within the fuel fired booster heater, replace the fuel fired booster heater combustion tube and evaporator assembly, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
B1D27-11	Heater Coolant Pump - Circuit short to ground	▪ Auxiliary coolant pump circuit short circuit to ground ▪ Pump failure	▪ Refer to the electrical circuit diagrams and check the auxiliary coolant pump circuit - Circuit AUX WATER PUMP - For short circuit to ground. Repair circuit as required, clear DTC and retest ▪ If fault persists, check and install a new auxiliary coolant pump as required. Clear DTC and retest
B1D27-12	Heater Coolant Pump - Circuit short to battery	▪ Auxiliary coolant pump circuit short circuit to power ▪ Pump failure	▪ Refer to the electrical circuit diagrams and check the auxiliary coolant pump circuit - Circuit AUX WATER PUMP - For short circuit to power. Repair circuit as required, clear DTC and retest ▪ If fault persists, check and install a new auxiliary coolant pump as required. Clear DTC and retest

B1D27-13	Heater Coolant Pump - Circuit open	▪ Auxiliary coolant pump circuit open circuit, high resistance ▪ Pump failure	▪ Refer to the electrical circuit diagrams and check the auxiliary coolant pump circuit - Circuit AUX WATER PUMP - For open circuit, high resistance. Repair circuit as required, clear DTC and retest ▪ If fault persists, check and install a new auxiliary coolant pump as required. Clear DTC and retest
B1D29-93	No Start, Even After Restart Attempt - No operation	▪ Fuel fired booster heater fuel pipe blocked/leaking ▪ Fuel fired booster heater exhaust restricted following driving through deep water or extreme off-road driving ▪ Fuel fired booster heater evaporator assembly contaminated with water or debris ingress following driving through deep water or extreme off-road driving ▪ Fuel fired booster heater internal failure	NOTES: ▪ This DTC is set when no flame is detected, even after re-start attempt ▪ If additional fuel fired booster heater related DTCs are also set, perform the relevant corrective action(s) first ▪ Check the fuel fired booster heater fuel pipe for blockages/leaks and replace as required ▪ If the fuel fired booster heater fuel pipe is replaced then, using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Fuel Priming (0x9009) ▪ Check the fuel fired booster heater exhaust for blockages and rectify as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater

			for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater ■ If a fuel fired booster heater fault persists, replace the fuel fired booster heater combustion tube and evaporator assembly, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A) ■ If the fault persists, install a new fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
B1D33-92	Flame Interruption During Normal Operation - Performance or incorrect operation	■ Fuel fired booster heater fuel pipe blocked/leaking ■ Fuel fired booster heater exhaust restricted following driving through deep water or extreme off-road driving ■ Fuel fired booster heater evaporator assembly contaminated with water or debris ingress	NOTES: ■ This DTC is set when no flame is detected, even after re-start attempt ■ If additional fuel fired booster heater related DTCs are also set, perform the relevant corrective action(s) first ■ Check the fuel fired booster heater fuel pipe for blockages/leaks and replace as required

		following driving through deep water or extreme off-road driving Fuel fired booster heater internal failure	- If the fuel fired booster heater fuel pipe is replaced then, using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Fuel Priming (0x9009) - Check the fuel fired booster heater exhaust for blockages and rectify as required - Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater - If a fuel fired booster heater fault persists, replace the fuel fired booster heater combustion tube and evaporator assembly, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A) - If the fault persists, install a new fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
B1D34-66	Heater In Lock Out Mode - Signal has too many transitions/events	Fuel fired booster heater fuel pipe	

				blocked/leaking	
				▪ Fuel fired booster heater exhaust restricted following driving through deep water or extreme off-road driving ▪ Fuel fired booster heater evaporator assembly contaminated with water or debris ingress following driving through deep water or extreme off-road driving ▪ Fuel fired booster heater internal failure	NOTES: ▪ This DTC is set when the fuel fired booster heater repeatedly fails to operate ▪ If additional fuel fired booster heater related DTCs are also set, perform the relevant corrective action(s) first ▪ Check the fuel fired booster heater fuel pipe for blockages/leaks and replace as required ▪ If the fuel fired booster heater fuel pipe is replaced then, using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Fuel Priming (0x9009) ▪ Check the fuel fired booster heater exhaust for blockages and rectify as required ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel

			fired booster heater ▪ If a fuel fired booster heater fault persists, replace the fuel fired booster heater combustion tube and evaporator assembly, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A) ▪ If the fault persists, install a new fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
P0530-11	A/C Refrigerant Pressure Sensor A Circuit - Circuit short to ground	▪ Refrigerant pressure sensor signal circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the refrigerant pressure sensor signal circuit for short circuit to ground
P0530-15	A/C Refrigerant Pressure Sensor A Circuit - Circuit short to battery or open	▪ Refrigerant pressure sensor signal circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the refrigerant pressure sensor signal circuit for short circuit to power, open circuit, high resistance
P06A1-11	Variable A/C Compressor Control Circuit Low - Circuit short to ground	▪ Air conditioning compressor solenoid negative circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the air conditioning compressor solenoid negative circuit for short circuit to ground
P06A1-12	Variable A/C Compressor Control Circuit Low - Circuit short to battery	▪ Air conditioning compressor solenoid negative circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the air conditioning compressor solenoid negative circuit for short circuit to power
P06A1-13	Variable A/C Compressor Control	▪ Air conditioning compressor	▪ Refer to the electrical circuit diagrams and check the air

	Circuit Low - Circuit open	solenoid negative circuit open circuit, high resistance	conditioning compressor solenoid negative circuit for open circuit, high resistance
P06A2-11	Variable A/C Compressor Control Circuit High - Circuit short to ground	Air conditioning compressor solenoid positive circuit short circuit to ground	Refer to the electrical circuit diagrams and check the air conditioning compressor solenoid positive circuit for short circuit to ground
P06A2-12	Variable A/C Compressor Control Circuit High - Circuit short to battery	Air conditioning compressor solenoid positive circuit short circuit to power	Refer to the electrical circuit diagrams and check the air conditioning compressor solenoid positive circuit for short circuit to power
P06A2-13	Variable A/C Compressor Control Circuit High - Circuit open	Air conditioning compressor solenoid positive circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the air conditioning compressor solenoid positive circuit for open circuit, high resistance
P1308-11	A/C Clutch Circuit - Circuit short to ground	Air conditioning compressor clutch circuit short circuit to ground	Refer to the electrical circuit diagrams and check the air conditioning compressor clutch circuit for short circuit to ground
P1308-12	A/C Clutch Circuit - Circuit short to battery	Air conditioning compressor clutch circuit short circuit to power	Refer to the electrical circuit diagrams and check the air conditioning compressor clutch circuit for short circuit to power
P1308-13	A/C Clutch Circuit - Circuit open	Air conditioning compressor clutch circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the air conditioning compressor clutch circuit for open circuit, high resistance
U0010-88	Medium Speed CAN Communication Bus -	Medium speed CAN bus	Using the manufacturer approved diagnostic system,

	Bus off	(comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance	perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	- Car configuration file mismatch with vehicle specification - Incorrect automatic temperature control module installed	- Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary - Install a new automatic temperature control module as necessary
U1000-00	Solid State Driver Protection Activated - Driver Disabled - No sub type information	Control module internal output driver is disabled due to a fault condition detected on an output circuit to protect the control module. Another DTC will be logged to show which specific output has the error	NOTE: The fault must be rectified on the output circuit before the DTC is cleared. Repeated logs of this diagnostic trouble code will result in unrecoverable control module shut down to prevent module damage Check the automatic temperature control module for output circuit DTCs and perform the relevant corrective action
U2101-56	Control Module Configuration Incompatible - Invalid / incomplete configuration	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary

U3000-01	Control Module - General electrical failure	Automatic temperature control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new automatic temperature control module
U3000-48	Control Module - Supervision software failure	Fuel fired booster heater internal failure	- Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A). This will operate the fuel fired booster heater for 15 minutes. Note that the routine should be started only when the engine is cool. The routine should be performed with the engine running and with the climate controls set to maximum heat and maximum fan. Repeat the routine again if smoke continues to be visible from the fuel fired booster heater exhaust pipe or from the area surrounding the fuel fired booster heater - If the fault persists, install a new fuel fired booster heater, clear the DTCs and perform routine - Auxiliary Heater (AHCM) Activation Test (0x900A)
U3000-87	Control Module - Missing message	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary
U3003-62	Battery Voltage - Signal compare failure	Mismatch between the voltage at the automatic temperature control module and the voltage value broadcast	Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111) - and compare it to battery voltage

		on the CAN bus	
--	--	----------------	--